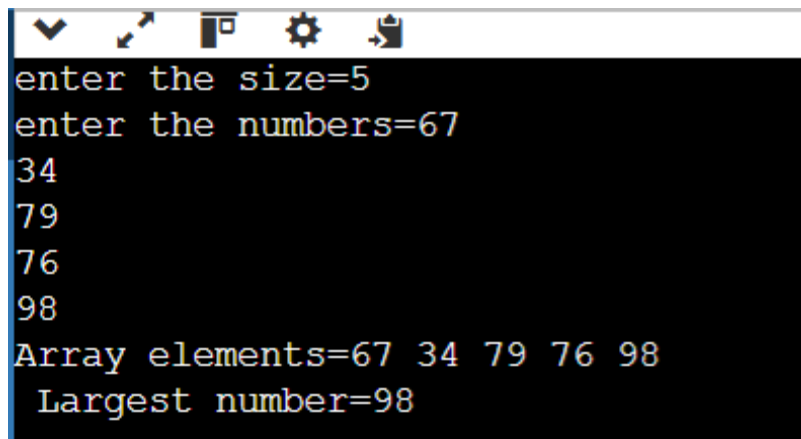


DATA STRUCTURE HW-3

1. Write a c program to store n elements in an array using dynamic memory allocation and find largest number.

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int n,i;
    printf("enter the size=");
    scanf("%d",&n);
    int *a=(int*)malloc(n*sizeof(int));
    printf("enter the numbers=");
    for(i=0; i<n; i++)
    {
        scanf("%d",&a[i]);
    }
    int max=a[0];
    for(i=0; i<n; i++)
    {
        if(a[i]>max)
        {
            max=a[i];
        }
    }
    printf("Array elements=");
    for(i=0; i<n; i++)
    {
        printf("%d ",a[i]);
    }
    printf("\n Largest number=%d",max);
```

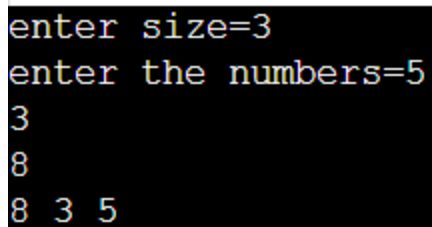
```
    free(a);  
    return 0;  
}
```

A screenshot of a terminal window with a black background and yellow text. The terminal shows the execution of a C program. It starts with a prompt 'enter the size=' followed by the input '5'. Then it prompts 'enter the numbers=' followed by the input '67'. The next three lines show inputs '34', '79', and '76'. The final line shows the output 'Array elements=67 34 79 76 98' and 'Largest number=98'.

```
enter the size=5  
enter the numbers=67  
34  
79  
76  
98  
Array elements=67 34 79 76 98  
Largest number=98
```

2. Print an array in reverse order using pointer.

```
#include <stdio.h>  
  
int main()  
{  
    int n,i;  
    int a[100];  
    int *p=a;  
    printf("enter size=");  
    scanf("%d",&n);  
    printf("enter the numbers=");  
    for(i=0;i<n;i++)  
    {  
        scanf("%d",&a[i]);  
    }  
    for(i=n-1;i>=0;i--)  
    {  
        printf("%d ",*(p+i));  
    }  
    return 0;  
}
```

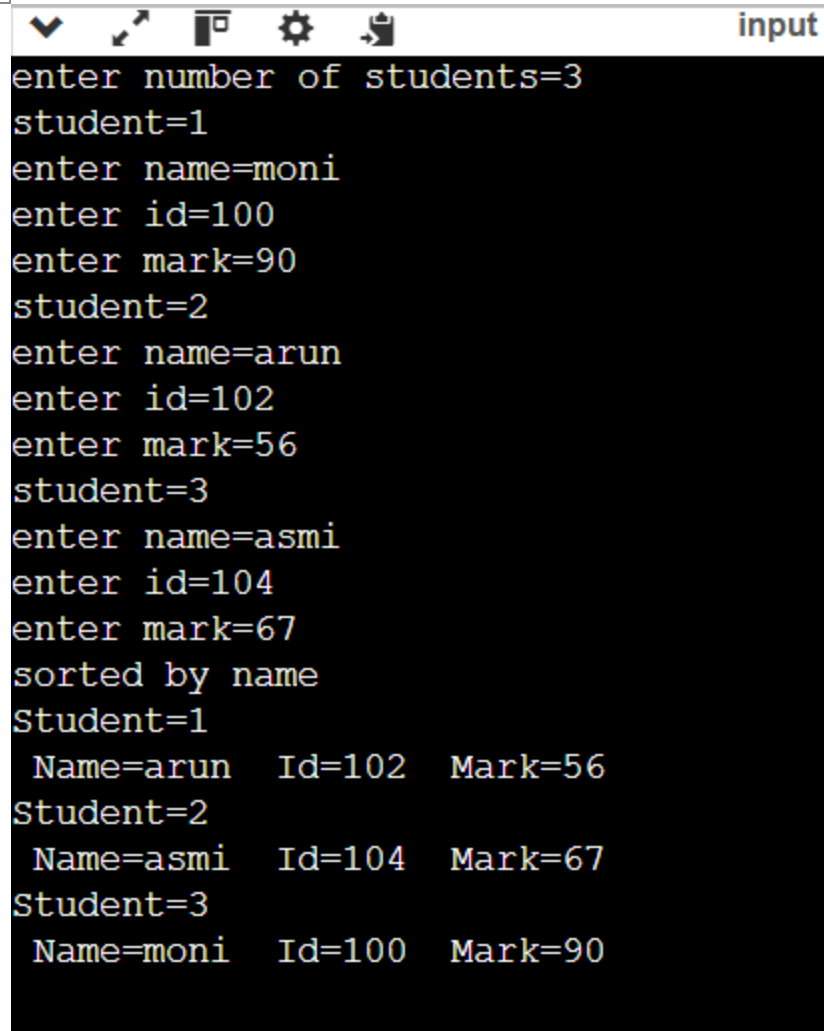


```
enter size=3
enter the numbers=5
3
8
8 3 5
```

3. Write a c program to store information of students using structure sort them by name.

```
#include <stdio.h>
#include <string.h>
struct student
{
    char name[100];
    int id;
    int mark;
};
int main()
{
    int n,i=0,j=0;
    printf("enter number of students=");
    scanf("%d",&n);
    struct student s[n];
    struct student temp;
    for(i=0;i<n;i++)
    {
        printf("student=%d\n",i+1);
        printf("enter name=");
        scanf("%s",s[i].name);
        printf("enter id=");
        scanf("%d",&s[i].id);
        printf("enter mark=");
        scanf("%d",&s[i].mark);
```

```
}  
for(i=0;i<n-1;i++)  
{  
    for(j=i+1;j<n;j++)  
    {  
        if(strcmp(s[i].name,s[j].name)>0)  
        {  
            temp=s[i];  
            s[i]=s[j];  
            s[j]=temp;  
        }  
    }  
}  
printf("sorted by name\n");  
for(i=0;i<n;i++)  
{  
    printf("Student=%d\n",i+1);  
    printf(" Name=%s ",s[i].name);  
    printf(" Id=%d ",s[i].id);  
    printf(" Mark=%d\n",s[i].mark);  
}  
}
```



```
input
enter number of students=3
student=1
enter name=moni
enter id=100
enter mark=90
student=2
enter name=arun
enter id=102
enter mark=56
student=3
enter name=asmi
enter id=104
enter mark=67
sorted by name
Student=1
  Name=arun   Id=102   Mark=56
Student=2
  Name=asmi   Id=104   Mark=67
Student=3
  Name=moni   Id=100   Mark=90
```

4. Write a c program to add 'n' distance given in inch-feet system using structure.

```
#include <stdio.h>

struct distance
{
    int feet;
    int inch;
};

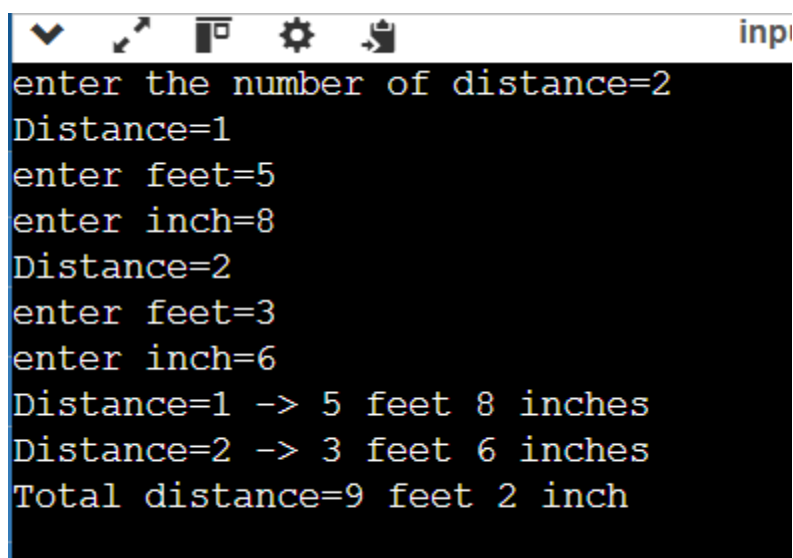
int main()
{
    int n,i;
    printf("enter the number of distance=");
    scanf("%d",&n);
    struct distance d[n];
```

```

struct distance sum={0,0};
for(i=0;i<n;i++)
{
    printf("Distance=%d\n",i+1);
    printf("enter feet=");
    scanf("%d",&d[i].feet);
    printf("enter inch=");
    scanf("%d",&d[i].inch);
}
for(i=0;i<n;i++)
{
    printf("Distance=%d -> ",i+1);
    printf("%d feet %d inches\n",d[i].feet,d[i].inch);
    sum.feet=sum.feet+d[i].feet;
    sum.inch=sum.inch+d[i].inch;
}
sum.feet=sum.feet+sum.inch/12;
sum.inch=sum.inch%12;;
printf("Total distance=%d feet %d inch\n",sum.feet,sum.inch);

return 0;
}

```



```

enter the number of distance=2
Distance=1
enter feet=5
enter inch=8
Distance=2
enter feet=3
enter inch=6
Distance=1 -> 5 feet 8 inches
Distance=2 -> 3 feet 6 inches
Total distance=9 feet 2 inch

```

5. Write a c program to add two complex number by passing structure to a function.

```
#include <stdio.h>

struct complexx
{
    int real;
    int img;
};

struct complexx add(struct complexx c1, struct complexx c2)
{
    struct complexx sum={0,0};
    sum.real=c1.real+c2.real;
    sum.img=c1.img+c2.img;
    return sum;
}

int main()
{
    int i;
    struct complexx c1,c2,result;

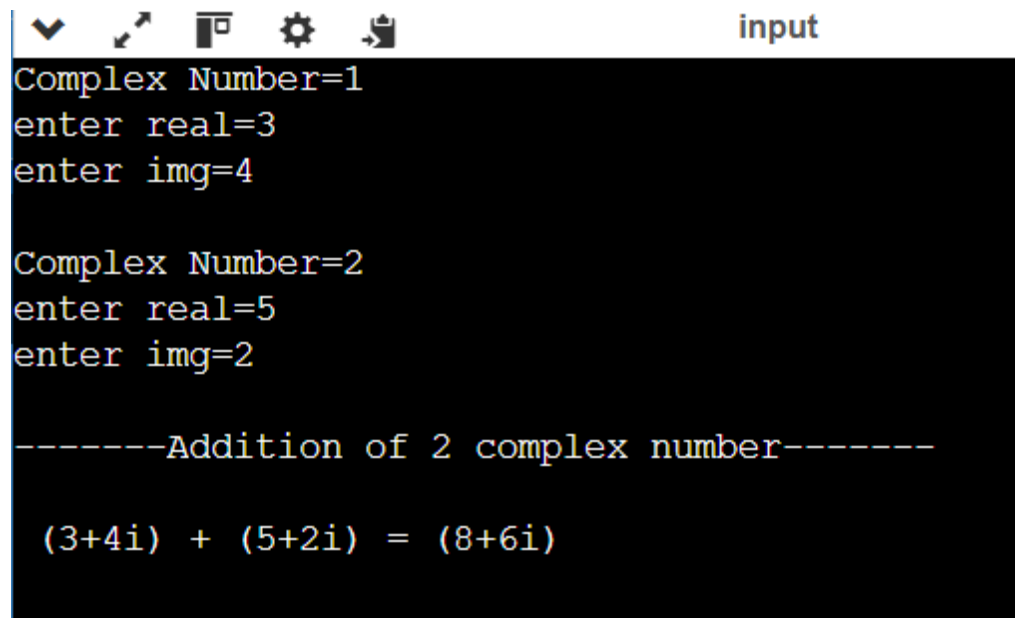
    printf("Complex Number=1\n");
    printf("enter real=");
    scanf("%d",&c1.real);
    printf("enter img=");
    scanf("%d",&c1.img);
    printf("\nComplex Number=2\n");
    printf("enter real=");
    scanf("%d",&c2.real);
    printf("enter img=");
    scanf("%d",&c2.img);
    result=add(c1,c2);
    printf("\n——Addition of 2 complex number——\n");
```

```

    printf("\n (%d+%di) + (%d+%di) = (%d+%di) \n",c1.real,c1.img,c2.real,c2.img,result
.real,result.img);

    return 0;
}

```



```

input
Complex Number=1
enter real=3
enter img=4

Complex Number=2
enter real=5
enter img=2

-----Addition of 2 complex number-----

(3+4i) + (5+2i) = (8+6i)

```

6. Write a c program to maintain inventory management system using structure.

```

#include <stdio.h>
#include <stdlib.h>
struct inventory
{
    char p_name[100];
    int id;
    int qty;
    int price;
};
int main()
{
    int i,n;
    printf("enter the no of product=");
    scanf("%d",&n);
    struct inventory *in;

```



```

in = (struct inventory*)malloc(n*sizeof(struct inventory));
for(i=0;i<n;i++)
{
    printf("Product=%d\n",i+1);
    printf("product name=");
    scanf("%s",in[i].p_name);
    printf("product id=");
    scanf("%d",&in[i].id);
    printf("product qty=");
    scanf("%d",&in[i].qty);
    printf("product price=");
    scanf("%d",&in[i].price);
}
int choice;
printf("1-add product\n2-display\n3-search product\n4-delete product\n5-update
qty\n6-exit");
do{
    printf("\nenter your choice=");
    scanf("%d",&choice);
    switch(choice)
    {
        case 1:
        {
            int m;
            printf("enter the number of product to be add=");
            scanf("%d",&m);
            in=(struct inventory*)realloc(in,(m+n)*sizeof(struct inventory));
            for(i=n;i<n+m;i++)
            {
                printf("Product=%d\n",i+1);
                printf("product name=");
                scanf("%s",in[i].p_name);
            }
        }
    }
}

```

```

printf("product id=");
scanf("%d",&in[i].id);
printf("product qty=");
scanf("%d",&in[i].qty);
printf("product price=");
scanf("%d",&in[i].price);
    }
    n=n+m;
    break;
}
case 2:
{
    printf("\n—————");
    printf("\n    inventory product    ");
    printf("\n—————\n");
    printf("\np_id\tp_name\tp_qty\tp_price");
    for(i=0;i<n;i++)
    {
        printf("\n%d\t%s\t%d\t%d",in[i].id,in[i].p_name,in[i].qty,in[i].price);
    }
    break;
}
case 3:
{
    int found=0;
    int search,pos=-1,i;
    printf("enter the product id to search=");
    scanf("%d",&search);
    for(i=0;i<n;i++)
    {
        if(in[i].id==search)
        {

```

```

        found=1;
        pos=i;
    }
}
if(found==1)
{
    printf("product found\n");
    printf("\np_id\tp_name\tp_qty\tp_price\t");

    printf("\n%d\t%s\t%d\t%d",in[pos].id,in[pos].p_name,in[pos].qty,in[pos].price);
}
else
{
    printf("product not found");
}
break;
}
case 4:
{
    int pos=-1;
    int delete,i;
    printf("enter the product id to delete=");
    scanf("%d",&delete);
    for(i=0;i<n;i++)
    {
        if(delete==in[i].id)
        {
            pos=i;
        }
    }
    for(i=pos;i<n;i++)
    {

```

```

        in[i]=in[i+1];
    }
    n--;
    break;
}
case 5:
{
    int new_qty,id;
    printf("enter the product id which to update the qty=");
    scanf("%d",&id);
    printf("enter the new_qty=");
    scanf("%d",&new_qty);
    int i,pos=-1;
    for(i=0;i<n;i++)
    {
        if(id==in[i].id)
        {
            pos=i;
        }
    }
    in[pos].qty=new_qty;
    break;
}
case 6:
{
    printf("\nexit\n");
    break;
}
}
}while(choice!=6);
free(in);
return 0;

```

```
}  
enter the no of product=2
```

```
Product=1
```

```
product name=tab
```

```
product id=1
```

```
product qty=7
```

```
product price=5999
```

```
Product=2
```

```
product name=lap
```

```
product id=2
```

```
product qty=6
```

```
product price=4300
```

```
1-add product
```

```
2-display
```

```
3-search product
```

```
4-delete product
```

```
5-update qty
```

```
6-exit
```

```
enter your choice=2
```

```
-----  
inventory product  
-----
```

```
p_id    p_name  p_qty  p_price
```

```
1       tab    7      5999
```

```
2       lap    6      4300
```

```
enter your choice=1
```

```
enter the number of product to be add=1
```

```
Product=3
```

```
product name=pods
```

```
product id=3
```

```
product qty=3
```

```
product price=8000
```

enter your choice=2

inventory product

p_id	p_name	p_qty	p_price
1	tab	7	5999
2	lap	6	4300
3	pods	3	8000

enter your choice=3

enter the product id to search=2

product found

p_id	p_name	p_qty	p_price
2	lap	6	4300

enter your choice=4

enter the product id to delete=3

enter your choice=2

inventory product

p_id	p_name	p_qty	p_price
1	tab	7	5999
2	lap	6	4300

enter your choice=5

enter the product id which to update the qty=1

enter the new_qty=9

```
enter your choice=2
```

```
-----  
inventory product  
-----
```

p_id	p_name	p_qty	p_price
1	tab	9	5999
2	lap	6	4300

```
enter your choice=6
```

```
exit
```

7. Write a c program for decimal to binary, octal and hexadecimal conversion.

```
#include <stdio.h>
```

```
#define max 100
```

```
int st[max], top=-1;
```

```
void push(int val)
```

```
{
```

```
    top++;
```

```
    st[top]=val;
```

```
}
```

```
int pop()
```

```
{
```

```
    int val;
```

```
    val=st[top];
```

```
    top--;
```

```
    return val;
```

```
}
```

```
int main()
```

```
{
```

```
    int decimal, rem, temp;
```

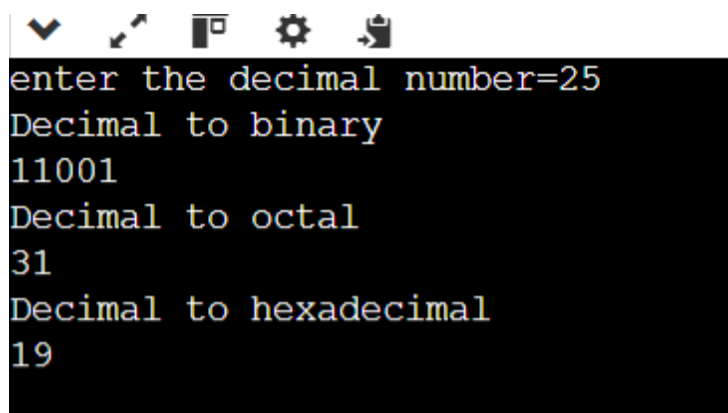
```
printf("enter the decimal number=");
scanf("%d",&decimal);
temp=decimal;
while(decimal>0)
{
    rem=decimal%2;
    push(rem);
    decimal=decimal/2;
}
printf("Decimal to binary\n");
while(top!=-1)
{
    printf("%d",pop());
}
decimal=temp;
while(decimal>0)
{
    rem=decimal%8;
    push(rem);
    decimal=decimal/8;
}
printf("\nDecimal to octal\n");
while(top!=-1)
{
    printf("%d",pop());
}
decimal=temp;
while(decimal>0)
{
    rem=decimal%16;
    push(rem);
    decimal=decimal/16;
```



```

    }
    printf("\nDecimal to hexadecimal\n");
    while(top!=-1)
    {
        int x;
        x=pop();
        if(x<10)
        {
            printf("%d",x);
        }
        else
        {
            printf("%c",x+55);
        }
    }
    return 0;
}

```



```

enter the decimal number=25
Decimal to binary
11001
Decimal to octal
31
Decimal to hexadecimal
19

```

8. Write a c program to calculate difference between two time periods.

```
#include <stdio.h>
```

```
struct timee
```

```

{
    int hour;
    int min;
    int sec;

```

```
};
```

```
int main()
```

```
{
```

```
    int i;
```

```
    struct timee t1,t2;
```

```
    printf("Time=1\n");
```

```
    printf("hour=");
```

```
    scanf("%d",&t1.hour);
```

```
    printf("minute=");
```

```
    scanf("%d",&t1.min);
```

```
    printf("secound=");
```

```
    scanf("%d",&t1.sec);
```

```
    printf("Time=2\n");
```

```
    printf("hour=");
```

```
    scanf("%d",&t2.hour);
```

```
    printf("minute=");
```

```
    scanf("%d",&t2.min);
```

```
    printf("secound=");
```

```
    scanf("%d",&t2.sec);
```

```
    struct timee dif;
```

```
    int sum1=0,sum2=0;
```

```
    sum1=t1.hour*3600+t1.min*60+t1.sec;
```

```
    sum2=t2.hour*3600+t2.min*60+t2.sec;
```

```
    if(sum1>sum2)
```

```
    {
```

```
        dif.sec=sum1-sum2;
```

```
    }
```

```
    else
```

```
    {
```

```
        dif.sec=sum2-sum1;
```

```
    }
```

```
    dif.hour=dif.sec/3600;
```

```

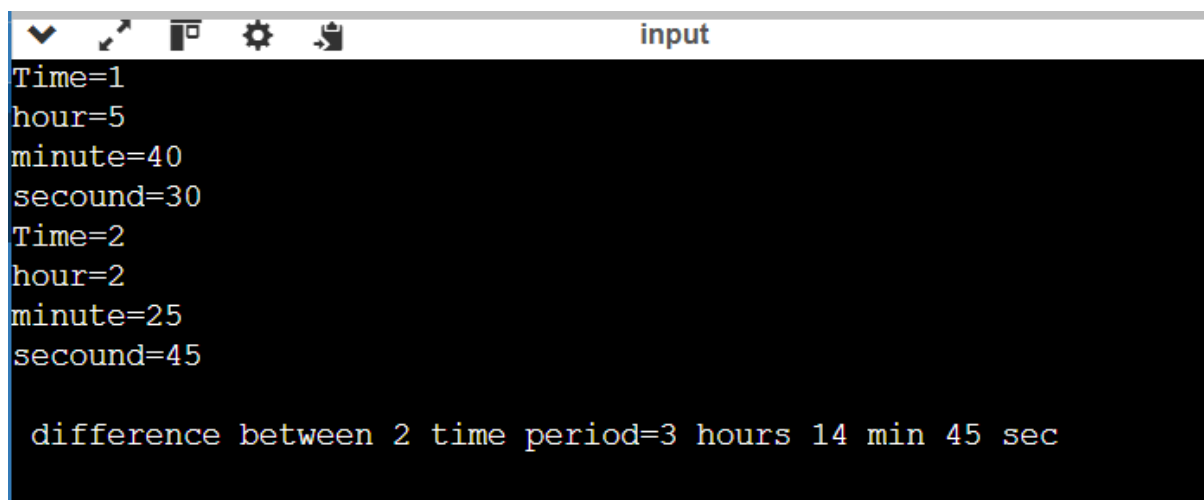
    dif.sec=dif.sec%3600;
    dif.min=dif.sec/60;
    dif.sec=dif.sec%60;

    printf("\n difference between 2 time period=%d hours %d min %d
sec",dif.hour,dif.min,dif.sec);

    return 0;

}

```



```

input
Time=1
hour=5
minute=40
secound=30
Time=2
hour=2
minute=25
secound=45

difference between 2 time period=3 hours 14 min 45 sec

```

9. Write a c program to implement an employee management system using structure.

```

#include <stdio.h>
#include <stdlib.h>
struct employee
{
    char name[100];
    int id;
    char dept[100];
    int salary;
};
int main()
{
    int n,i;
    printf("enter the number of employee=");

```

```

scanf("%d",&n);
struct employee *e;
e=(struct employee*)malloc(n*sizeof(struct employee));
for(i=0;i<n;i++)
{
    printf("\nemployee=%d\n",i+1);
    printf("\nname=");
    scanf("%s",e[i].name);
    printf("\nid=");
    scanf("%d",&e[i].id);
    printf("department=");
    scanf("%s",e[i].dept);
    printf("salary=");
    scanf("%d",&e[i].salary);
}
int choice;
printf("\n1-add\n2-display\n3-search\n4-count\n5-delete\n6-exit\n");
do{
    printf("\nenter choice=");
    scanf("%d",&choice);
    switch(choice)
    {
        case 1:
        {
            int m;
            printf("enter no of employee to add=");
            scanf("%d",&m);
            e=(struct employee*)realloc(e,(m+n)*(sizeof(struct employee)));
            for(i=n;i<(m+n);i++)
            {
                printf("\nemployee=%d\n",i+1);
                printf("\nname=");

```

```

scanf("%s",e[i].name);
printf("\nid=");
scanf("%d",&e[i].id);
printf("department=");
scanf("%s",e[i].dept);
printf("salary=");
scanf("%d",&e[i].salary);
    }
    n=n+m;
    break;
}
case 2:
{
    printf("\n_____");
    printf("\n   employee management system ");
    printf("\n_____");
    printf("\ne_id\te_name\te_dept\te_salary");
    for(i=0;i<n;i++)
    {
        printf("\n%d\t%s\t%s\t%d",e[i].id,e[i].name,e[i].dept,e[i].salary);
    }
    break;
}
case 3:
{
    int search;
    int found=0,pos=-1;
    printf("enter id to search=");
    scanf("%d",e[i].id);
    for(i=0;i<n;i++)
    {
        if(search==e[i].id)

```

```

        {
            found=1;
            pos=i;
        }
    }
    if(found==1)
    {
        printf("employee found\n");
        printf("\ne_id\te_name\te_dept\te_salary");
        for(i=0;i<n;i++)
        {
            printf("\n%d\t%s\t%s\t%d",e[pos].id,e[pos].name,e[pos].dept,e[pos].salary);
        }
    }
    else
    {
        printf("Not found\n");
    }
    break;
}
case 4:
{
    printf("Number of employee=%d\n",n);
    break;
}
case 5:

{
    int pos=-1;
    int delete,i;
    printf("enter the id to delete=");
    scanf("%d",&delete);

```

```
        for(i=0;i<n;i++)
        {
            if(delete==e[i].id)
            {
                pos=i;
            }
        }
        for(i=pos;i<n;i++)
        {
            e[i]=e[i+1];
        }
        n--;
        break;

    }

    case 6:
    {
        printf("exit\n");
        break;
    }

}

}while(choice!=6);
return 0;
}
```

enter the number of employee=2

employee=1

name=moni

id=100

department=it

salary=8000

employee=2

name=yashu

id=102

department=cse

salary=7000

1-add

2-display

3-search

4-count

5-delete

6-exit

enter choice=2

employee management system

e_id	e_name	e_dept	e_salary
------	--------	--------	----------

100	moni	it	8000
-----	------	----	------

102	yashu	cse	7000
-----	-------	-----	------

enter choice=1

enter no of employee to add=1

employee=3

name=vijay


```
enter choice=1
enter no of employee to add=1
employee=3
name=vijay
id=104
department=cse
salary=4566
```

```
enter choice=2
```

```
-----
                employee management system
-----
```

e_id	e_name	e_dept	e_salary
100	moni	it	8000
102	yashu	cse	7000
104	vijay	cse	4566

```
enter choice=3
enter id to search=2
Not found
```

```
enter choice=3
enter id to search=102
employee found
```

e_id	e_name	e_dept	e_salary
102	yashu	cse	7000

```
enter choice=4
Number of employee=3
```

```
enter choice=4
Number of employee=3
```

```
enter choice=5
enter the id to delete=100
```

```
enter choice=2
```

```
-----  
employee management system  
-----
```

```
e_id    e_name  e_dept  e_salary
```

```
102     yashu   cse     7000
```

```
104     vijay  cse     4566
```

```
enter choice=6
```

```
exit
```

10. Write a c program to calculate area and perimeter of a rectangle and triangle using function pointers.

```
#include <stdio.h>
```

```
float rec_area(float l,float br)
```

```
{
```

```
    return l*br;
```

```
}
```

```
float rec_perimeter(float l,float br)
```

```
{
```

```
    return 2*(l+br);
```

```
}
```

```
float tri_area(float a,float b,float c)
```

```
{
```

```
    return (a+b+c)/2;
```

```
}
```

```
float tri_perimeter(float a,float b,float c)
```

```
{
```

```
    return a+b+c;
```

```
}
```

